

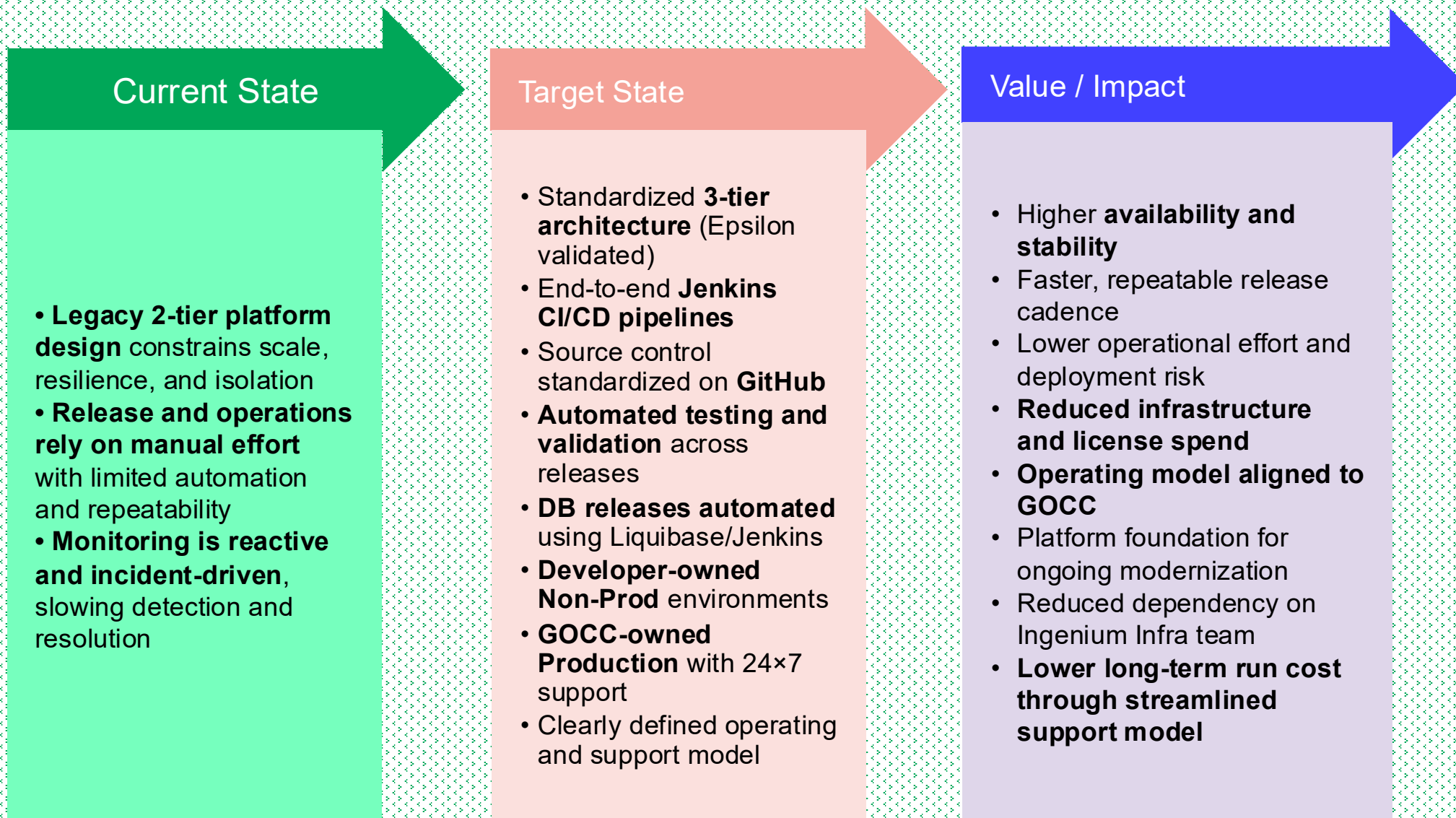
ETS Ingenium – Updates

Executive Update | Balaji Ravi

April 21, 2026



Ingenium – Modernization Roadmap



Ingenium — Quarterly Update & Plan

Q1- Update

1. Modernization architecture design finalization.
2. CICD design for the ingenium services.
3. Ingenium KT – High level plan on task segregation has been completed

- Design document
- 3 tier architecture design
- CICD pipeline design for Ingenium services
- Source code migration from Version Manager -> Github migration plan
- Epsilon – Project plan

Q2- Progress

1. Implement the 3-tier architecture in POT/DEV subscription.
2. CICD pipeline creation for the various Ingenium components
3. Version Manager to Github migration
4. Ingenium KT – Path to GOCC
5. BAU activities automation & Improvements and Documentations

- Benefits & Outcome
- Establishes a standardized, scalable, and cost-optimized architecture with improved performance and resilience and proven code migration to Github.
- Enables faster, predictable, and lower-risk releases through end-to-end automation
- Ensures a smooth transition to GOCC with clear ownership, 24x7 support, and reduced infra dependency.
- Reduces manual operational effort while enabling consistent support and gradual Ingenium Infra resource reduction.

Ingenium — Quarterly Update & Plan

Q3 -2026 ~ Q2-2027

- 1. Rollout Q2 Outcomes to Production & CI/CD Enablement for Production Releases
- 2. Complete GitHub Adoption & Standardization
- 3. Complete Ingenium KT & GOCC Transition
- 4. BAU Automation, Stabilization & Optimization Activity Plan
- 5. MQ implementation POT & Batch transition to GBO
- 6. COBOL standardization.

- Delivers a stable, production-ready Ingenium platform with predictable, low-risk deployments and improved operational confidence.
- Establishes a single, governed source of truth, improving code quality, collaboration, auditability, and long-term maintainability.
- Enables full GOCC-led production ownership with clear accountability, 24×7 support, and reduced dependency on Ingenium Infra resources.
- Reduces manual effort and operational risk while lowering long-term run cost through standardized, repeatable BAU processes.
- Improves integration resilience and batch reliability, while enabling GOCC ownership of batch operations with performance monitoring

Key Initiatives	Q3-2026			Q4-2026			Q1-2027			Q2-2027
	Jul-26	Aug-26	Sep-26	Oct-26	Nov-26	Dec-26	Jan-27	Feb-27	Mar-27	Apr-27
Modernization 3 tier Architecture - Epsilon	3-Tier Rollout to Production									
Version Manger to Github Migration							Version Manager Exit			
CICD Automation	Build, Deployment & Test Automation									
Ingenium KT - Path to GOCC	2-Tier Architecture					3-Tier Architecture				

Modernization Roadmap: Key Initiatives and Business Benefits

Pattern	Actual price	Discounted price
Total cost for old pattern	¥806,890,538	¥242,067,161
Total cost for new pattern	¥625,942,448	¥187,782,734
Savings from new pattern	¥180,948,090	¥54,284,427

~22% savings

Modernization – Architecture

(4 Resource)

Ingenium – CICD

(3 Resource)

Ingenium KT – Path to GOCC

(7 BAU +2 Release)

Business Benefits (Value/Impact)

1) Licensing savings (annual / recurring)

- Annual licensing savings: \$180K~\$250K (approx. ¥30M~¥40M) per year
- Value/Impact: Recurring savings from the new licensing pattern under the target architecture, realized once the 3-tier architecture is rolled out to Production

2) Workforce optimization (phased, dependency-based)

- Phase 1 (end of transition): 16 → 7 by end of Q4 2026
Driver: transition completion + initial operating model stabilization
- Phase 2 (post 3-tier architecture standardization): Workforce reduction from 7 → 4 by end of Q3 2027, subject to stable BAU operations under the standardized 3-tier model.

Assumption/Dependency: Workforce optimization savings depend on GOCC’s availability to complete knowledge transfer (KT). If KT completion is delayed, the planned savings may be delayed and may not be fully realized within the target timeline

Q3 Plan – Required Support for 3-Tier Production Rollout

What we need from DEV— and what you get in return

What we're asking DEV to do

- **Sanity-check** the new **3-tier environment** end-to-end (app, services, data), bug fix support during testing
- **Confirm configuration + integrations** (secrets, endpoints, connectivity, dependencies)
- **Support deploy + rollback readiness**(runbooks, smoke tests, rollback steps validated)
- **Use GitHub effectively**
 - Keep work in **PRs**, enforce **reviews**, keep branches short-lived
 - Use **Issues/Boards** for tracking and clear ownership
 - Maintain **README/runbooks** and versioned configuration where applicable
- **Use Jenkins CI/CD to Non-Prod**
 - Standard pipeline to **build** → **test** → **package** → **deploy** into non-prod
 - Ensure **pipeline gates** (automated checks) before promotion
 - Produce **audit trail** (who/what/when) for deployments

What DEV gets (benefits)

- **Fewer environment-specific surprises** and less last-minute debugging
- **Faster, predictable releases** through repeatable pipelines
- **Earlier feedback loop** via automated validation and non-prod parity
- **Better visibility** (logs/metrics/artifacts captured by pipeline) for quicker root-cause analysis
- **Reusable deployment patterns** that reduce ongoing support effort and rework
- **Increased automation reduces manual effort, enabling workforce optimization over time.**

Q3 Plan – Required Support for 3-Tier Production Rollout

What we need from AQA — and what you get in return

What we're asking AQA to do

- **Extensive validation across SIT / UAT / PS** against the **3-tier architecture**
 - Functional + integration coverage across **Web / App / DB**
 - **Batch validation** (end-to-end runs, schedule/dependencies, failure handling)
 - **Performance checks** (batch throughput, response times, stability under load)
- **Validate non-functional readiness**
 - Regression coverage, data integrity, and cross-component behavior
 - Confirm key **SLAs / SLOs** for batch + online flows where applicable
- **Collaborate with Ingenium Infra to expand automation in Jenkins**
 - Increase automated **smoke / regression / batch verification** coverage
 - Parameterize tests to run reliably across environments and datasets
 - Add **pipeline gates** (pass/fail criteria) so issues are caught early
- **Partner with Development during testing for faster bug fix cycles**
 - Tight triage loop (repro steps, logs, evidence, severity)
 - Validate fixes quickly and keep testing moving without rework churn

What AQA gets (benefits)

- **More stable SIT/UAT/PS cycles** with fewer reruns caused by environment drift
- **Earlier defect detection** and clearer root-cause signals (logs/metrics + pipeline evidence)
- **Less manual repetitive work** as Jenkins automation coverage increases
- **Faster turnaround on defects** through tighter Dev–AQA collaboration
- **More predictable release outcomes** (reduced last-minute surprises)
- **Reusable automation assets** that scale across releases, regions, and environments
- **Increased automation reduces manual effort, enabling workforce optimization over time**

Thank You

Ingenium – One of the Key Benefit

New Pattern Licensing Cost

Key Points:

- 1. Licensing cost shown below is a **one-time cost** (VCS license cost not included).
- 2. Cost is calculated across all environments (SIT/UAT, PS, PROD, Dev, POT).
- 3. New pattern reduces overall license volumes and provides opportunities for reuse.

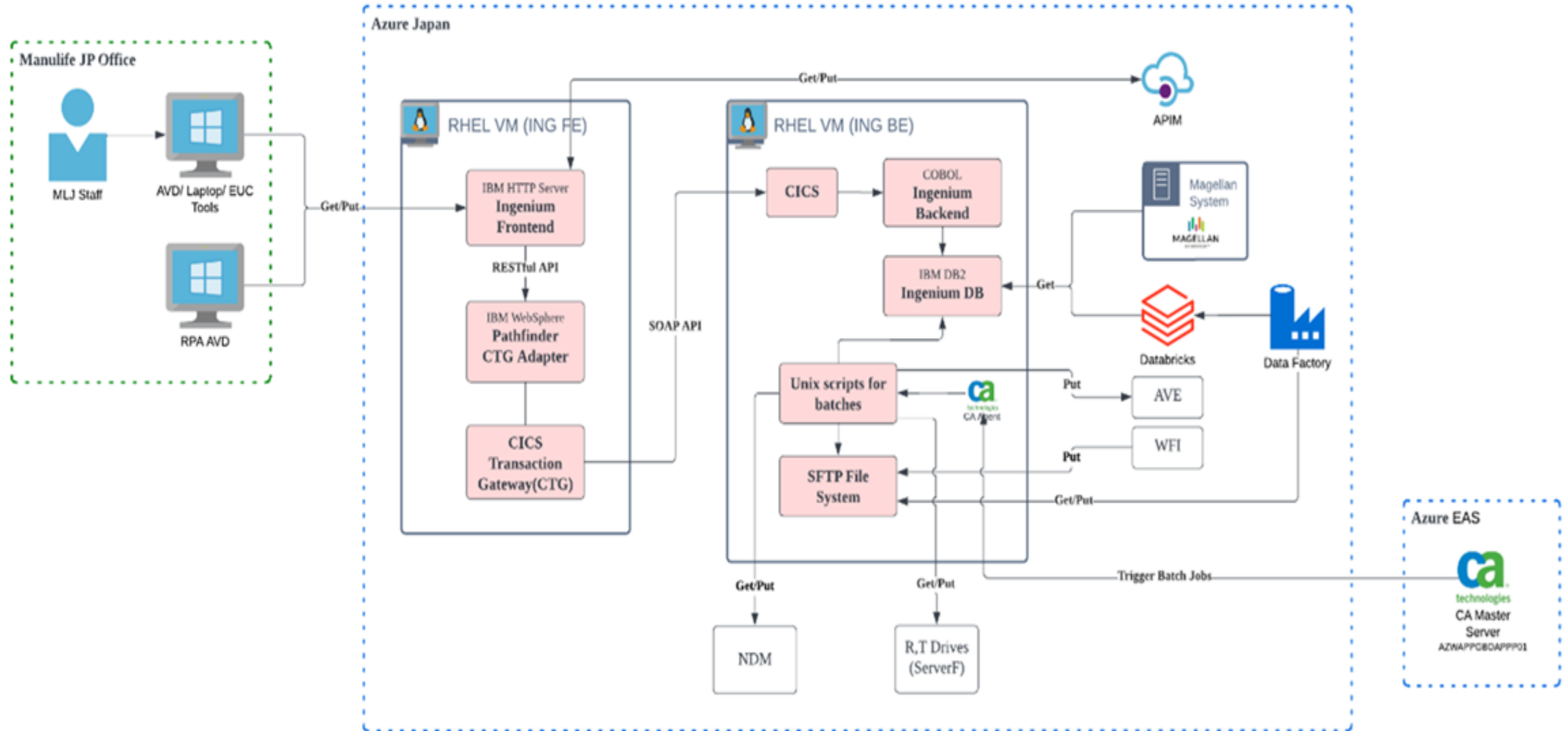
Benefits:

- 1. The **new architecture pattern** significantly reduces overall licensing costs.
- 2. Even *before* reuse, the new pattern gives **~22% cost reduction**.
- 3. With license reuse included, additional **~¥155M+ cost advantage** is unlocked.
- 4. Discounted savings at ~70% bring effective cost reduction.

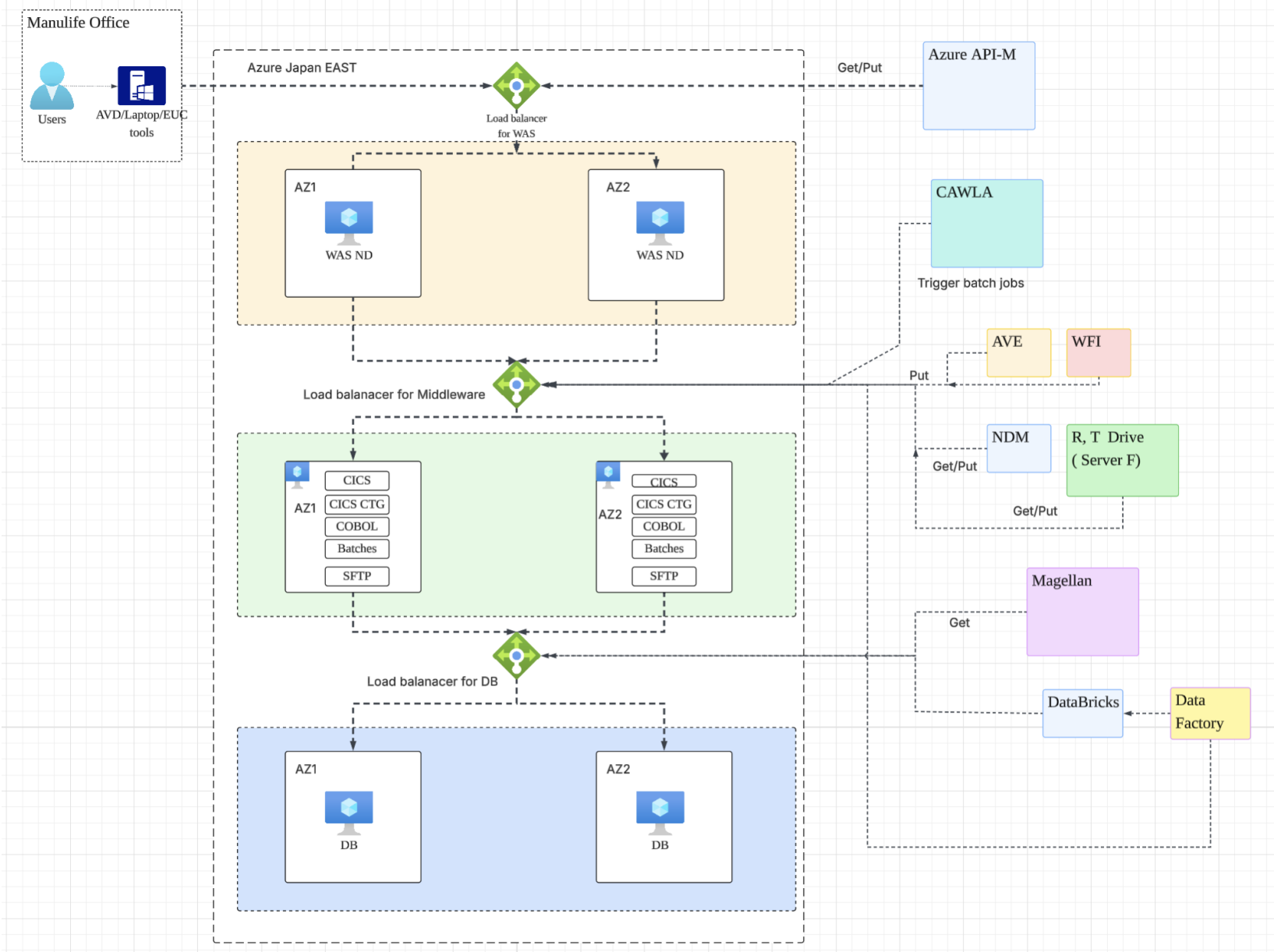
Expense Category	Cost Saving Initiatives	Owner
IT Related expense	Review system design in Development, Testing and PS team to decrease the system maintenance cost	Nagayasu, Ravi, Manish, Yogi & Birger
IT Related expense	License/maintenance cost	Vendor Management

Category	Total Cost (Actual Price)	After 70% Discount
Total Cost – Old Pattern	¥806,890,538	¥242,067,161
Total Cost – New Pattern	¥625,942,448	¥187,782,734
Savings by Moving to New Pattern	¥180,948,090 (22%)	¥54,284,427
License Reuse Benefit	¥155,690,481	¥46,707,144

Ingenium Current Architecture



Ingenium To be Architecture



Ingenium – Licensing cost for old pattern

				Test(SIT/UAT)		PS		PROD		Dev		POT	
	Price per Unit(CAD)	Unit Per Core	Price per Unit(Yen)	CPU Quantity-license	Cost in Yen	CPU Quantity-license	Cost in Yen	CPU Quantity- license	Cost in Yen	CPU Quantity-license	Cost in Yen	CPU Quantity-license	Cost in Yen
DB2 SE	16,154.00	1	1,819,758.92	16	29,116,142.77					0	0	0	0
DB2 AE	82,983.00	1	9,348,090.55		0.00	16	149,569,448.78	16	149,569,448.78	0	0	0	0
TX Series	405	70	45,623.52	16	51,098,343.91	16	51,098,343.91	16	51,098,343.91	0	0	0	0
CTG	224	70	25,233.75	16	28,261,800.09	16	28,261,800.09	16	28,261,800.09	0	0	0	0
WAS	21,258.00	1	2,394,727.94	16	38,315,647.09	16	38,315,647.09	16	38,315,647.09	0	0	0	0
COBOL User	5,858	1	659,868.20	17	11,217,759.35	5	3,299,340.99	5	3,299,340.99	0	0	0	0
COBOL Core	19,296	1	2,173,707.33	16	34,779,317.25	16	34,779,317.25	16	34,779,317.25	0	0	0	0
					192,789,010.46	305,323,898.10		305,323,898.10					

Total cost for old pattern803,436,806.67

70% discount241,031,042.00

Ingenium – Licensing cost for new pattern

				Test(SIT/UAT)		PS		PROD		Dev		POT	
	Price per Unit(CAD)	Unit Per Core	Price per Unit(Yen)	CPU Quantity-license	Cost in Yen	CPU Quantity-license	Cost in Yen	CPU Quantity-license	Cost in Yen	CPU Quantity-license	Cost in Yen	CPU Quantity-license	Cost in Yen
DB2 SE	16,154.00	1	1,819,758.92	16	29,116,142.77					2	3639517.846	3	5459276.77
DB2 AE	82,983.00	1	9,348,090.55			17	158,917,539.33	17	158917539.3				
TX Series	405	70	45,623.52	8	25,549,171.96	8	25,549,171.96	8	25549171.96	2	6,387,292.99	2	6,387,292.99
CTG	224	70	25,233.75	8	14,130,900.04	8	14,130,900.04	8	14130900.04	2	3,532,725.01	2	3,532,725.01
WAS	21,258.00	1	2,394,727.94	4	9,578,911.77	4	9,578,911.77	4	9578911.771	2	4,789,455.89	4	9,578,911.77
COBOL User	5,858	1	659,868.20	17	11,217,759.35	5	3,299,340.99	5	3299340.986	5	3,299,340.99	5	3,299,340.99
COBOL Core	19,296	1	2,173,707.33	8	17,389,658.63	16	34,779,317.25	16	34779317.25	2	4,347,414.66	0	0.00
Total				106,982,544.52		246,255,181.34		246,255,181.34		25,995,747.37		28,257,547.53	

Total cost for new pattern	625,488,654.57	70 % discount	8,477,264.26
70% discount	187,646,596.37		

Ingenium – Licensing cost for reuse

				Test(SIT/UAT)		PS		PROD		Dev		POT	
	Price per Unit(CAD)	Unit Per Core	Price per Unit(Yen)	CPU Quantity-license	Cost in Yen	CPU Quantity-license	Cost in Yen	CPU Quantity-license	Cost in Yen	CPU Quantity-license	Cost in Yen	CPU Quantity-license	Cost in Yen
DB2 SE	16,154.00	1	1,819,758.92	0	0.00	0	0.00	0	0.00	2	3,639,517.85	-3	-5,459,276.77
DB2 AE	82,983.00	1	9,348,090.55	0	0.00	-1	-9,348,090.55	-1	-9,348,090.55	0	0.00	0	0.00
TX Series	405	70	45,623.52	8	25,549,171.96	8	25,549,171.96	8	25,549,171.96	-2	-6,387,292.99	-2	-6,387,292.99
CTG	224	70	25,233.75	8	14,130,900.04	8	14,130,900.04	8	14,130,900.04	-2	-3,532,725.01	-2	-3,532,725.01
WAS	21,258.00	1	2,394,727.94	12	28,736,735.31	12	28,736,735.31	12	28,736,735.31	-2	-4,789,455.89	-4	-9,578,911.77
COBOL User	6,993	1	787,784.16	0	0.00	0	0.00	0	0.00	-5	-3,938,920.80	-5	-3,938,920.80
COBOL Core	19,296	1	2,173,707.33	8	17,389,658.63	0	0.00	0	0.00	-2	-4,347,414.66	0	0.00

85,806,465.94

59,068,716.77

59,068,716.77

-19,356,291.49

-28,897,127.34

Total - Delta 155,690,480.64

70% discount 46707144.19

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Overview of key modernization initiatives and outcomes

